Dr. Sabina Jehan Haque

James Van Loo Independent Postdoctoral Research Fellow
University of Michigan, Department of Mathematics

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Employment

University of Michigan Ann Arbor — Jul 2024 - present
Van Loo Independent Postdoctoral Research Fellow

Education

Harvard University — Sep 2018 - May 2024
PhD in Systems Biology
Advisor: Jeremy Gunawardena
Dissertation title: Graph-theoretic approaches to biochemical reaction networks

Middlebury College — Sep 2014 - May 2018
B.A. in Mathematics (high honors),
Molecular Biology & Biochemistry

Research interests: algebraic graph theory, dynamical systems theory, applied algebraic geometry, chemical reaction network theory, continuous-time Markov processes, non-equilibrium biophysics, fast-slow dynamics and model reduction, structural identifiability

Publications

**Manuscript available upon request*

Submitted

1. ***Haque S. J.**, Eisenberg, M. Effects of infinite parameter limits on the structural identifiability of linear compartment models. Submitted to *SIAM J. Appl. Algebra Geom.* (2026).

Published

2. **Haque, S. J.**, Satriano, M., Sorea, M. & Yu, P. Y. The Disguised Toric Locus and Affine Equivalence of Reaction Networks. *SIAM J. Appl. Dyn. Syst.* **22**, 1423–1444 (2023).
3. Chavez, A., Tuttle M, Pruitt B. W., Ewen-Campen B., Chari R., Ter-Ovanesyan D., **Haque S. J.** *et al.* Comparison of Cas9 activators in multiple species. *Nat. Meth.* **13**, 563–567 (2016).

In Preparation

4. *__Haque S. J.__, Nam K.-M., Gunawardena J. A graph construction for analysing the parametric asymptotics of Markov processes (2026).
5. *__Haque S. J.__, Cetiner U., Gunawardena J. Anomalous behaviour of the Steinberg signature for detecting departure from thermodynamic equilibrium. (2026).

Fellowships and Awards

James Van Loo Independent Postdoctoral Fellowship 2024 - 2027
Michigan Center for Applied and Interdisciplinary Mathematics (MCAIM), University of Michigan Ann Arbor
• $4K travel and research funding each year for 3 years of postdoctoral fellowship

SMB Travel Grant for CRNT 2026 Workshop in Hawaii 2026
• Lodging accommodated for week-long workshop in chemical reaction networks

Junior Scientists Board Research Grant Challenge ($10K funding) 2024 - 2025
Center for Complex Particle Systems (COMPASS), University of Michigan Ann Arbor

Lynch Foundation PhD Fellowship 2019 - 2021, 2023 - 2024
Department of Systems Biology, Harvard University
• 100% salary and tuition support for 3 years of PhD

NSF-Simons QBio PhD Fellowship 2021 - 2022
NSF-Simons Center for Mathematical & Statistical Analysis of Biology
• 50% salary and tuition support for 1 year, $4K travel funding

Selected Talks

Exploring the effect of parametric model reduction on the structural identifiability of linear compartment models Jan 15 2027
Special session on “Identifiability and Indistinguishability in Dynamical Systems”, Joint Mathematics Meetings 2027, Chicago, IL

Singular parametric limits on graphs, dynamical systems, and algebraic varieties Jan 15 2027
Special session on “Algebraic, Combinatoric, and Stochastic Approaches in Mathematical Biology”, Joint Mathematics Meetings 2027, Chicago, IL

A graph construction for analysing the parametric asymptotics of Markov processes Jan 14 2027
Special session on “Analysis of Interaction Network Models and Its Application to Mathematical Biology”, Joint Mathematics Meetings 2027, Chicago, IL

Discrete structure, continuous dynamics: Graph-theoretic approaches and singular parametric limits in biochemical reaction networks Sep 25 2026
Applied and Interdisciplinary Mathematics (AIM) Seminar, University of Michigan, Ann Arbor MI

Exploring the effect of parametric model reduction on the structural identifiability of linear compartment models — Jul 13 2026
14th European Conference on Mathematical & Theoretical Biology, University of Graz, Graz, Austria

Exploring the effect of parametric model reduction on the structural identifiability of linear compartment models — May 18 2026
Chemical Reaction Networks Workshop in Hawai'i 2026, University of Hawai'i at Mānoa, Honolulu, HI

Graph-theoretic approaches to biochemical reaction networks and structural identifiability — Feb 05 2026
Applied Algebra Group, Department of Mathematics, University of Copenhagen, Copenhagen, Denmark

Graph-theoretic and algebraic geometric approaches to biochemical reaction networks — Nov 11 2025
Mathematics of Reaction Networks (MoRN) Seminar, virtual

Graph-theoretic and algebraic geometric approaches to biochemical reaction networks — Oct 03 2025
Mathematics Seminar, Algebraic Systems Biology Group, MPI-CBG, Dresden, Germany

A graph construction for analysing the parametric asymptotics of Markov processes — Jul 12 2025
2025 SIAM Conference on Applied Algebraic Geometry, 4th Madison Workshop on Mathematics of Reaction Networks, Madison, WI

Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. — Jan 12 2024
National Institute for Theoretical and Mathematical Biology, Chicago IL

Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. — Oct 17 2023
Barcelona Collaboratorium for Modelling and Predictive Biology, UPF, Barcelona Spain

A graph-theoretic approach to Markov processes with applications in biochemical reaction networks. — Oct 03 2023
Mathematics Department Colloquium, Middlebury College, Middlebury VT

Following the energy: graph-theoretic models of broken detailed balance with biochemical applications. — Apr 25 2023
NSF-Simons QBio Seminar, Harvard University, Cambridge MA

Graph-theoretic models of detecting broken detailed balance in molecular information processing. — Jul 11 2022
2022 SIAM Annual Meeting MS 26: Trends and New Results in Deterministic Models of Biochemical Interaction Networks, Pittsburgh PA

Teaching and mentoring

University of Michigan Ann Arbor
Instructor of record:
MATH 404: Intermediate Differential Equations — Winter 2025
(Introduction to Nonlinear Dynamics & Chaos) — Fall 2025, Fall 2026

MATH 464: Inverse Problems — Winter 2027
- Designed undergraduate topics course in Foundations of Chemical Reaction Networks

Undergraduate students mentored:
Joe Bogdan, University of Michigan — Summer 2025
- Project title: Toric geometry of chemical reaction networks (CRNs)
- Independent research study on network operations and toric dynamical systems; poster presentation at SIAM Student Chapter Symposium 2025

Daniel Le (Colby College) and George Simmons (University of Michigan) — Summer 2026
- Project title: Computing Gröbner fans to understand parameter geometry in mathematical biology
- Math REU project focused on Gröbner fan algorithm development

Harvard University
Teaching fellow:
AM 50: Introduction to Applied Mathematics — Spring 2020

Service

Conferences and seminars organized

Co-organizer for JMM Special Session (accepted) — Jan 2027
Joint Mathematics Meetings, Chicago IL
- Session title: “Algebraic, Combinatoric, and Stochastic Approaches in Mathematical Biology”
- Co-organizers: Dr. Ruby Kim, Dr. Naghmeh Akhavan

Organizer for MCAIM Colloquium — 2026-present
University of Michigan, Ann Arbor MI
- Invited Dr. Jinsu Kim for a visit and a colloquium talk at Michigan Center for Applied and Interdisciplinary Mathematics

University and departmental service

Math Corps summer instructor for high school students — 2026-present
University of Michigan, Ann Arbor MI
- Developed and taught 5-week class “What is graph theory” for 9th grade students

Intercollegiate Math Modeling Challenge (IM²C) judge Nov 2025
University of Michigan, Ann Arbor MI

Association for Women in Mathematics (AWM) peer mentor 2024-present
University of Michigan, Ann Arbor MI
- Run peer mentoring groups for graduate and undergraduate students at University of Michigan

Public engagement

Math Club speaker Oct 2024
Department of Mathematics, University of Michigan, Ann Arbor MI

Guest speaker for summer class on Physics of Biological Function Jun 2023
Wentworth Institute of Technology, Boston MA

Cambridge Science Festival volunteer Oct 2022
Cambridge, MA

"What is Systems Biology?" outreach event creator/coordinator Apr 2019
Cambridge Science Festival, Cambridge MA
- Developed an interactive public event for the Cambridge Science Festival with hands-on activities illustrating concepts from systems biology

Science communication

Cellular mathematics 2022-present
Public-facing Substack on mathematics and biology

Graph Theory 101 Jun-Aug 2021
Science in the News: Networks: https://sitn.hms.harvard.edu/flash/2021/graph-theory-101/

Professional activities

Referee, SIAM Journal of Life Sciences (SIALS) 2026-present

Member, Society of Industrial and Applied Mathematics (SIAM) 2025-present
Member, Society of Mathematical Biology (SMB) 2026-present